

1 Definitions, Classification of PDEs

1.1 Big O, Classification

Big O: $f(t) = \mathcal{O}(g(t))$ as $t \rightarrow t_0$ if $\exists \varepsilon > 0, C > 0$ s.t. $\forall |t - t_0| < \varepsilon, |f(t)| \leq C|g(t)|$
 $\Leftrightarrow \lim_{t \rightarrow t_0} [f(t)/g(t)]$ bdd ps: If not say $t \rightarrow t_0$, assume $t \rightarrow 0$

More on Big O: a. $F(x) = G(x) + \mathcal{O}(H(x))$ if $F(x) - G(x) = \mathcal{O}(H(x))$

- b. $F(x) = \mathcal{O}(G(x)) + \mathcal{O}(H(x))$ if $F(x) = \mathcal{O}(\max\{|G(x)|, |H(x)|\})$
 c. $F(x, y) = \mathcal{O}(m(x)) + \mathcal{O}(y^n)$ [if Given $G(x) = \mathcal{O}(x^{\frac{m}{n}}) \Rightarrow F(x, G(x)) = \mathcal{O}(x^m)$], 且另一项同理

Classification: For $a u_{xx} + b u_{xy} + c u_{yy} + d u_x + e u_y + f u + g = 0$ ($a, b, c \neq (0, 0, 0)$)

- Hyperbolic:** $b^2 - 4ac > 0$ Admits non-trivial plane wave solutions (Propagation)
- Parabolic:** $b^2 - 4ac = 0$ and we can get $2cd = be$ or $2ae = bd$
- Elliptic:** $b^2 - 4ac < 0$ No non-trivial plane wave solutions (No Propagation)

Remark: 对变量 x, y 进行缩放后不会改变 PDE 类型

1.2 Normal

Weighted 2-Norm: $\|\mathbf{u}\|_{2, \Delta x} = \sqrt{(\Delta x)^d \|\mathbf{u}\|_2}$ where d s.t. $\mathbf{u} : \mathbb{R}^d \rightarrow \mathbb{R}^n$

- For $d = 1$ and $\Delta x = \frac{D}{N}$, then: $\forall i, u_i = \mathcal{O}((\Delta x)^p) \Rightarrow \|\mathbf{u}\|_{2, \Delta x} = \mathcal{O}((\Delta x)^p)$
- For A symmetric: $\|A^{-1} \mathbf{w}\|_2 \leq \frac{1}{|\lambda_1|} \|\mathbf{w}\|_2$ $0 < |\lambda_1| \leq \dots \leq |\lambda_n|$

Property: If $\|\mathbf{u}\|_2 \leq \|\mathbf{w}\|_2$, then $\|\mathbf{u}\|_{2, \Delta x} \leq \|\mathbf{w}\|_{2, \Delta x}$

New Norm: The norm $\|\mathbf{u}\| := \max_n \|\mathbf{u}^n\|_{2, \Delta x}$ where $\mathbf{u}^n = [u_1^n, u_2^n, \dots, u_{M-1}^n]^\top$
 Which means: Infinity norm in time, weighted 2-norm in space. 主要用于双变量 PDE

For $\tau_m^n = \mathcal{O}((\Delta x)^\alpha) + \mathcal{O}((\Delta t)^b)$, $\Delta x = \frac{b-\Delta t}{M-\Delta t}$, $\Delta t = \frac{b}{N}$, assume $\Delta t = K(\Delta x)^\beta$, then:
 $\max_n \|\mathbf{x}^n\|_{2, \Delta x} = \mathcal{O}((\Delta x)^{\min\{\alpha, b\}})$

1.3 Linear Algebra

Definite: Positive Definite: $\lambda_i > 0, \forall i$ Negative Definite: $\lambda_i < 0, \forall i$

Symmetric: If A symmetric, then $A = Q \Lambda Q^\top$ where Q orthogonal, Λ diagonal with eigenvalues.

Normal: Real matrix A is normal if $A A^\top = A^\top A$ The following are properties of it:

- A is orthogonally diagonalizable, i.e. $A = Q \Lambda Q^\top$ where Q orthogonal, Λ diagonal with eigenvalues.
- Exists an eigenvectors of A : $\{\mathbf{v}_i\}$ s.t. $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij}$ and Symmetric $\Rightarrow$ Normal

2 Global/Local Error, Convergence, Consistency, Stability

Exact & Approx. Sol: U_n : approx. n -th time-step sol; u_n : exact sol at n -th time-step.

Global Error $(e_n, \mathbf{e}(\Delta t))$: $e_n = U_n - u(t = n\Delta t)$ $\mathbf{e}(\Delta t) = [e_0, e_1, \dots, e_N]^\top$

Convergence: A numerical method is convergent if $\lim_{\Delta t \rightarrow 0} \|\mathbf{e}(\Delta t)\| = 0$

If $\lim_{\Delta t \rightarrow 0} \frac{\|\mathbf{e}(\Delta t)\|}{(\Delta t)^p} = C$ (i.e. $\|\mathbf{e}(\Delta t)\| = \mathcal{O}((\Delta t)^p)$) $\Rightarrow$ p -th order accurate

Local Truncation Error (τ_n) : 数值分析的离散迭代公式, 将 U_n 替换为 u_n 后的误差 (LHS - RHS).

- e.g. For $\frac{U_{n+1} - U_n}{\Delta t} = f(U_n)$, $\tau_{n+1} = \frac{U_{n+1} - u_{n+1}}{\Delta t} - f(u_n)$ $\tau_0 = 0$
- e.g. 我们可以把线性数值方法写成 $Au = b$, A matrix, $u = [U_1, \dots, U_{N-1}]^\top$.

Then: $\tau = A u_{\text{exact}} - b$ where $u_{\text{exact}} = [u_1, \dots, u_{N-1}]^\top$, $\tau = [\tau_1, \dots, \tau_{N-1}]^\top$

ps: $Ae = -\tau$ where $e = u - u_{\text{exact}}$ Define $\tau(\Delta t) = [\tau_1, \dots, \tau_{N-1}]^\top$

Consistency: If $\lim_{\Delta t \rightarrow 0} \|\tau(\Delta t)\| = 0$, the method is consistent.

Linear Equation: Goal: Finally change ODE/PDE into $Au = b$ where A is matrix, $u = [U_1, \dots, U_{N-1}]^\top$

Stability: A numerical method is stable if $\exists \Delta x_0$ s.t. For the Linear System $Au = b$:

- $A(\Delta x)$ is invertible $\forall \Delta x < \Delta x_0$
- $\|A(\Delta x)^{-1} \mathbf{w}\| \leq C \|\mathbf{w}\|$ for all $\mathbf{w} \in \mathbb{R}^{N-1}$

Property: Consistent + Stable $\Rightarrow$ Convergent (same norm!) and $\|\mathbf{e}(\Delta t)\| = \mathcal{O}(\|\tau(\Delta t)\|)$

3 Intro & Method of Lines

Taylor Series: $f(t + \Delta t) = f(t) + f'(t)\Delta t + \frac{1}{2}f''(t)(\Delta t)^2 + \dots + \mathcal{O}((\Delta t)^N)$ ps: $u_{n+1} = u(n\Delta t + \Delta t)$

$f(t) = f(a) + f'(a)(t-a) + \frac{f''(a)}{2!}(t-a)^2 + \dots + \frac{f^{(N-1)}(a)}{(N-1)!}(t-a)^{N-1} + \mathcal{O}((t-a)^N)$

Euler Methods: Forward: $\frac{du}{dt}|_n \Delta t \approx \frac{U_{n+1} - U_n}{\Delta t}$; Backward: $\frac{du}{dt}|_n \Delta t \approx \frac{U_n - U_{n-1}}{\Delta t}$

Leapfrog: $\frac{du}{dt}|_n \Delta t \approx \frac{U_{n+1} - U_{n-1}}{2\Delta t}$ **Trapezoidal:** $\frac{1}{2} \left[\frac{du}{dt}|_n \Delta t + \frac{du}{dt}|_{n+1} \Delta t \right] \approx \frac{U_{n+1} - U_n}{\Delta t}$

Centred Difference: $\frac{d^2 u}{dt^2}|_n \Delta t \approx \frac{U_{n+1} - 2U_n + U_{n-1}}{(\Delta t)^2}$

Method of Lines: Using $U_m(t) \approx u(a + m\Delta x, t)$, $U_m^n \approx u(a + m\Delta x, n\Delta t)$ a 为空间起点, 不重要

- For 1D:** a. 先对空间 x 离散化, 使用 U_m 表达, 此时 $\frac{d}{dt} \mathbf{U} = \dots$ where $\mathbf{U} = [U_1(t), U_2(t), \dots, U_{M-1}(t)]^\top$
 b. 对时间 t 使用 ODE 离散化, 使用 U_m^n 表达, 得到 Linear Equation.

- Example: Crank-Nicolson for Heat Eq. $u_t = \kappa u_{xx}$: 对 x 用 centred difference; 对 t 用 trapezoidal:

$$\frac{U_{m+1}^n - U_m^n}{\Delta t} = \frac{\kappa}{2} \left(\frac{U_{m+1}^{n-1} - 2U_m^{n-1} + U_{m-1}^{n-1}}{(\Delta x)^2} + \frac{U_{m+1}^n - 2U_m^n + U_{m-1}^n}{(\Delta x)^2} \right)$$

$$\Rightarrow -r U_{m-1}^{n+1} + (2 + 2r) U_m^{n+1} - r U_{m+1}^{n+1} = r U_{m-1}^n + (2 - 2r) U_m^n + r U_{m+1}^n, \quad r = \frac{\kappa \Delta t}{(\Delta x)^2}$$

Solve it: By iteration: e.g. $A \mathbf{U}^{n+1} = B \mathbf{U}^n$, where $\mathbf{U}^n = [U_1^n, U_2^n, \dots, U_{M-1}^n]^\top$.

Truncation for MoL: 对于 τ_m^n , 离散 x 后 U_m 替换为 u_m ; 对于 τ_m^n , 再离散 t 后 U_m^n 替换为 u_m^n .

e.g. For Crank-Nicolson above: $\tau_m = \frac{\partial u_m}{\partial t} - \kappa \frac{u_{m+1} - 2u_m + u_{m-1}}{(\Delta x)^2}$

e.g. For Crank-Nicolson above: $\tau_m^n = \frac{u_{m+1}^n - u_m^n}{\Delta t} - \frac{\kappa}{2}(\dots)$

4 Lax-Richtmyer Stability, von Neumann Analysis

Abs. Stability for ODEs: For ODEs $\frac{du}{dt} = A u$, u vector. A 1-step method for t (i.e. $U^{n+1} = B U^n$)

It's absolutely stable if $\|B \mathbf{w}\|_2 \leq \|\mathbf{w}\|_2$ for all $\mathbf{w} \in \mathbb{R}^{N-1}$

Property: If B normal, then absolutely stable $\Leftrightarrow |\lambda_i(B)| \leq 1, \forall i$

Property: Absolute stability $\Rightarrow$ Lax-Richtmyer stability in 2-norm and weighted 2-norm

Lax-Richtmyer Stability: For PDE which discrete form $\frac{1}{\Delta t} U^{n+1} = \frac{1}{\Delta t} B U^n + F^n$, B fixed, F can change.

It's Lax-Richtmyer stable if $\|B^n \mathbf{w}\| \leq C \|\mathbf{w}\|$ for all $\mathbf{w} \in \mathbb{R}^{N-1}$, $n \leq \frac{1}{\Delta t} = N$

Property: If consistent with $\max_n \|\mathbf{x}^n\|_{2, \Delta x} = \mathcal{O}((\Delta x)^\beta)$ and L-R stable $\|B^n \mathbf{w}\|_{2, \Delta x} \leq C \|\mathbf{w}\|_{2, \Delta x}$,

then, it converges with $\max_n \|\mathbf{e}^n\|_{2, \Delta x} = \mathcal{O}((\Delta x)^\beta)$ Remark: Always Assume $\Delta t = K(\Delta x)^\alpha$

von Neumann Analysis: PDE: 1. periodic BCs in spatial x , length D . 2. 满足 L-R 中 PDE 框架, 3. $F^n = 0$

$\Rightarrow$ Let $v_{k,m} = \exp(2\pi i k \frac{m \Delta x}{D})$ for $m = 0, 1, \dots, M-1$. Then vector $\mathbf{v}_k$ 为 B 的正交特征向量.

* By $U_m^n = \exp(2\pi i k \frac{m \Delta x}{D})$, $U_m^{n+1} = \alpha_k U_m^n$, 可得特征值(放大因子) α_k 表达式 (i.e. $B \mathbf{v}_k = \alpha_k \mathbf{v}_k$).

$\Rightarrow$ If all α_k for $k = 0, 1, \dots, M-1$ can be found $\Rightarrow B$ is normal and then:

If $|\alpha_k| \leq 1, \forall k \Rightarrow B$ is absolutely stable and L-R stable in weighted 2-norm

Remark: 假设 $\Delta t = K(\Delta x)^\alpha$ 会得 K, α 的限制条件, $\|$ For $F^n \neq 0$, 讨论 $F^n = 0$ 足够.

5 Boundary Condition

2nd ODE: For $u_x x = \beta(x)$ with $x \in (a, b)$, there are three types of BCs:

- Two Dirichlet:** $u(x=a) = c$ $u(x=b) = d$
- One Dirichlet, One Neumann:** $u(x=a) = c$ $u_x(x=b) = d$ or $u_x(x=a) = c$ $u(x=b) = d$
- One Neumann, One Constraint:** $u_x(x=a) = c$ or $\int_a^b u(x) dx = d$

Heat PDE: For $u_t = \kappa u_{xx}$ with $x \in (a, b)$, there are three types of BCs: * Basic IC: $u(x, t_0) = P(x)$

- Two Dirichlet:** $u(x=a, t) = c$ $u(x=b, t) = d$
- One Dirichlet, One Neumann:** $u(x=a, t) = c$ $u_x(x=b, t) = d$ or $u_x(x=a, t) = c$ $u(x=b, t) = d$
- Two Neumann:** $u_x(x=a, t) = c$ $u_x(x=b, t) = d$

Remark: For $u_x(x, a, t) = c \Rightarrow$ describe $\frac{u_m^n - u_0^n}{\Delta x} = c$ Boundary condition itself can effect the accuracy

For $u_t(x, t_0) = P(x) \Rightarrow$ describe $\frac{u_m^n - u_0^n}{\Delta t} = P(a + m\Delta x)$

6 One Time Solve Two Variables PDEs

Elliptic Equation: $L u = F(x_1, \dots, x_d)$ is an elliptic PDE if L is an elliptic operator.

Theorem: $L = \sum_{r=1}^d \sum s = 1^d A_{r,s} \frac{\partial^2}{\partial x_r \partial x_s} + \sum_{n=1}^d B_n \frac{\partial}{\partial x_n} + C$ is elliptic operator if:

A is symmetric and A is positive/negative definite ($\lambda_i > 0 | \lambda_i < 0$).

Poisson Equation: $\Delta u = F(x_1, \dots, x_d)$ is a Poisson equation and it's elliptic. with BCs:

Dirichlet: $u = u_D$ on $\partial \Omega$; Neumann: $\Delta u \cdot \hat{\mathbf{n}} = g$ on $\partial \Omega$ where $\int_{\Omega} F = \int_{\partial \Omega} g$ and $\int_{\Omega} u = \int_{\partial \Omega} u_D$.

e.g. For 2D: $u_{xx} + u_{yy} = F(x, y)$ with BCs: $u(x, y) = u_D$ on $\partial \Omega$

$\rightarrow$ 5-point stencil: $\frac{U_{m-1,p} + U_{m+1,p} + U_{m,p-1} + U_{m,p+1} - 4U_{m,p}}{(\Delta x)^2} = F(a + m\Delta x, a + p\Delta x)$

$\rightarrow$ Using $\mathbf{u} = [U_{1,1}, U_{2,1}, \dots, U_{M-1,1}, U_{1,2}, \dots, U_{M-1,M-1}]^\top$ and

$\mathbf{b}_{m+(p-1)}(M-1) = F(a + m\Delta x, a + p\Delta x)$ then $A \mathbf{u} = \mathbf{b}$ where A is a symmetric matrix.

$\rightarrow$ Eigenvalue/vectors of A : $\mathbf{v}_{k,l} = [v_{k,l,1,1}, v_{k,l,1,2}, \dots, v_{k,l,1,2}, \dots, v_{k,l,M-1,M-1}]^\top$
 $v_{k,l,m,p} = \sin(\pi k m \Delta x) \sin(\pi l p \Delta x)$, $\lambda_{k,l} = \frac{2}{(\Delta x)^2} [\cos(\pi k \Delta x) + \cos(\pi l \Delta x) - 2] < 0$, $\lambda_{\min} = \lambda_{1,1}$

Remark: 这一节都是用同一个 Δx 离散 x 和 y 的, 和前面 t, x 分别不同.

Consistency: If $\lim_{\Delta x \rightarrow 0} \frac{\|\tau(\Delta x)\|}{\Delta x} = 0$, the method is consistent.

Convergence: If $\|\mathbf{e}(\Delta x)\| = \mathcal{O}((\Delta x)^p)$, the method is p -th order accurate.

Stability: A numerical method is stable if for the linear system $A \mathbf{u} = \mathbf{b}$:

- $A(\Delta x)$ is invertible $\forall \Delta x < \Delta x_0$
- $\|A(\Delta x)^{-1} \mathbf{w}\| \leq C \|\mathbf{w}\|$ for all $\mathbf{w} \in \mathbb{R}^{N-1}$

Property: If A normal, then $\|A^{-1} \mathbf{w}\|_2 \leq \frac{1}{|\lambda_1|} \|\mathbf{w}\|_2$ where $0 < |\lambda_1| \leq \dots \leq |\lambda_{N-1}|$ are eigenvalues of A .

7 Spectral Methods

Inner Product Space for Function: $\langle \phi_i, \phi_j \rangle = \int_a^b \phi_i \phi_j w dx$ ps: $w(x)$ is weight function.

For Fourier Series: $w(x) = 1, a = -\pi, b = \pi$: $\sin(n x), \cos(m x)$ are orthogonal basis.

Fourier Series: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n x) + b_n \sin(n x) = \sum_{n=-\infty}^{\infty} A_k e^{i k x}$ where

$$\rightarrow a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(n x) dx \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(n x) dx.$$

$$\rightarrow k > 0: A_k = \frac{1}{2} (a_k - i b_k); \quad k < 0: A_k = \frac{1}{2} (a_{-k} + i b_{-k}); \quad k = 0: A_0 = \frac{a_0}{2}. \quad \text{For real func: } A_{-k} = \overline{A_k}$$

Property: $A_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-i k x} dx$ and $f'(x) = \sum_{k=-\infty}^{\infty} (i k) A_k e^{i k x}$

More on Fourier Series: Assume f is periodic with period 2π , then:

- The Fourier Transform: $\mathcal{F}\{f(x)\} = \hat{f}(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-i k x} dx$
- The inverse Fourier Transform: $\mathcal{F}^{-1}\{\hat{f}(k)\} = f(x) = \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{i k x}$

Change $L \rightarrow 2\pi$ period: If $f(x)$ has period $L \Rightarrow g(x) = f(\frac{L}{2\pi} x)$ has 2π period.

Discrete Fourier Transform (DFT): If $x_n = n\Delta x$ equally-spaced with $n = 0, 1, \dots, N-1$; $\Delta x = \frac{2\pi}{N}$:

- Fourier Transform:** $\mathcal{F}\{f\} = \hat{f}_k = \frac{2\pi}{N} \sum_{n=0}^{N-1} f(x_n) e^{-i k x_n}$ $k = -\frac{N}{2} + 1, \dots, \frac{N}{2}$

- Inverse FT:** $\mathcal{F}^{-1}\{\hat{f}\} = f(x) = \frac{1}{2\pi} \sum_{k=-N/2+1}^{N/2} \hat{f}_k e^{i k x_n} = \frac{1}{2\pi} \sum_{k=-N/2}^{N/2} \hat{f}_k e^{i k x_n}$

where $\hat{f}_k = \hat{f}_k$ for $k = -\frac{N}{2} + 1, \dots, \frac{N}{2} - 1$; $\hat{f}_{-N/2} = \hat{f}_{N/2} = \frac{1}{2} \hat{f}_{N/2}$.

Property: $\hat{f}'_k = (i k) \hat{f}_k \Leftrightarrow \mathcal{F}\{f'\} = (i k) \mathcal{F}\{f\}$

Remark: Quickly compute DFT: Fast Fourier Transform (FFT)

Using Fourier to solve PDE: For PDE with periodic BCs

- Take Fourier Transform in PDE (in space) e.g. $u_t = u_{xx}$ $\rightarrow \frac{d}{dt} \hat{u}_k = -k^2 \hat{u}_k$
- Take Fourier Transform in IC (in space) e.g. $u(x, 0) = f(x) \rightarrow \hat{u}_k(t=0) = \hat{f}_k$
- Using Transformed PDE and IC $\rightarrow$ a. Directly Solve ODE. b. numerical method for ODE. $\rightarrow \hat{u}_k(t)$
- Inverse Fourier Transform to get $u(x_n, t)$ 对 x_n 点 t 的估计或精确解.

Ideal of Interpolants: 给定 $(x_n, f(x_n))$, 构造插值函数 $p(x)$ s.t. $p(x_n) = f(x_n) \Rightarrow$ 用 $p'(x_n)$ 估计 $f'(x_n)$.

Spectral Collocation: Given $(x_n, f(x_n))$, f has period 2π , construct Fourier Interpolant $p(x)$ s.t. $p(x_n) = f(x_n)$

- DFT: $\hat{f}_k = \frac{2\pi}{N} \sum_{n=0}^{N-1} f(x_n) e^{-i k x_n} \Rightarrow p(x) = \frac{1}{2\pi} \sum_{k=-N/2}^{N/2} \hat{f}_k e^{i k x}$
- (Method I) (即 FFT, 无法得到微分算子的矩阵) Direct differentiate $p(x)$, and $f'(x_n) \approx p'(x_n)$.

- (Method II) Assume $p(x) = \sum_{m=0}^{N-1} f(x_m) P_\delta(x - x_m)$ where $P_\delta(x_j - x_m) = \delta_{j-m}$
 $\Rightarrow P_\delta(x - x_m) = \frac{1}{2\pi} \sum_{k=-N/2}^{N/2} \hat{P}_{\delta k} e^{i k x} = \frac{1}{2\pi} \sum_{k=-N/2}^{N/2} e^{i k(x-x_m)} \text{ and } P_\delta(x) = \frac{\sin(\pi x / \Delta x)}{(2\pi / \Delta x) \tan(x/2)}$

$$\Rightarrow f'(x_n) \approx p'(x_n) = \sum_{m=0}^{N-1} f(x_m) P'_\delta(x_n - x_m) \quad \text{and } P'_\delta(x_n - x_m) = \begin{cases} 0 & j \equiv 0 \pmod{N} \\ \frac{1}{2} (-1)^j \cot(\frac{i h}{2}) & j \not\equiv 0 \pmod{N} \end{cases}$$

$\Rightarrow$ Can be written in matrix form: $\mathbf{w} = D \mathbf{v}$ 其中 $\mathbf{w}$ 是 $f'(x_n)$ 的估计向量, $\mathbf{v}$ 是 $f(x_n)$ 的向量 and $f^{(n)} \Leftrightarrow D^n$

Chebyshev Spectral Method: For non-periodic BCs. Assume points are $x_n = \cos(\frac{n\pi}{N})$ for $n =$

11 Appendix

11.1 Other

Divergence Theorem: $\int_{\Omega} \nabla \cdot \mathbf{F} dV = \int_{\partial\Omega} \mathbf{F} \cdot \hat{\mathbf{n}} dS$ $\hat{\mathbf{n}}$ outward normal. $\nabla^2 = \nabla \cdot \nabla = \Delta$
How to improve Approximation Accuracy?: a. Using more neighboring points. (more terms from Taylor expansion) b. Make $\Delta t, \Delta x$ smaller. c. For Periodic BCs, use Fourier Transform.